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Zinc-based deep eutectic solvent – An efficient carbonic anhydrase mimic for CO_2 hydration and conversion

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ABSTRACT

Carbonic anhydrase (CA) has demonstrated great potential to mitigate CO_2 emissions by enzymatic conversion of CO_2, while its commercial implementation is limited by the inherent shortcomings of CA, such as thermal and chemical instability, high sensitivity to the environment, and high cost. To overcome the drawbacks of CA and develop advanced technologies for mitigating CO_2 emission, for the first time, mimetic CA (ZnHisGly) with the characteristics of natural CA was designed and synthesized, i.e., zinc-based deep eutectic solvent (DES), to boost CO_2 hydration and conversion. The mimetic CA exhibited facile synthesis, high stability, and low cost as the characteristics of DES, and showed better catalytic performance compared to the currently reported mimetic CA. Particularly, its catalytic performance increased greatly with increasing pH and temperatures, which provides promising prospects in the industrial applications of DES-based CA mimics.

1. Introduction

The emission of CO_2 has been increasing at an alarming rate due to the rapid growth in energy demand and excessive combustion of fossil fuels in recent years [1]. Developing separation and purification technologies as well as those of CO_2 conversion to mitigate CO_2 emissions is of great importance to reduce the negative impact of CO_2 on climate change (e.g., global warming and glaciers melting), and also to be beneficial for sustainable development and circular economy [2]. Currently, different technologies have been developed for CO_2 conversion, including chemical [3], photocatalytic [4], electrochemical [5], and biological transformation [6]. Among them, biological conversion by using carbonic anhydrase (CA) to efficiently convert CO_2 to HCO_3^- is a promising one, owing to its high specificity and selectivity under the mild condition and presenting environmentally friendly nature [7–9].

Despite the excellent performance of CA for CO_2 conversion, the practical applications of CA are limited owing to the challenges of thermal and chemical instability, high sensitivity to pH and temperatures, and high cost [10]. For instance, the biological activity of CA could drop to zero after few minutes when the temperature is above 65 °C [11], and both denaturation and peptide hydrolysis would occur in a highly alkaline environment [12]. Besides, CA is costly due to the difficulties of cell culture and enzyme purification in production. To alleviate the problems of CA, immobilization of CA to solid materials, such as epoxy-functionalized magnetic polymer microspheres [13], activated paramagnetic Fe_3O_4 nanoparticles [14], etc., has been proposed. Although the immobilized CA improved thethermal stability and recycling compared to the free CA, this strategy still suffers from enzyme leaking and the unavoidable reduction of enzymatic activity.

Recently, mimetic CA as a promising alternative is being developed for overcoming the shortcomings of natural CA, and it can achieve high stability, controllable structures, and low cost [15]. Since the active-site of natural CA (Scheme 1) is mainly composited by forming the coordination of zinc ion with three histidine molecules, azoles or their derivatives are widely used as alternative ligands of histidine, and the generalized principle is followed to synthesize mimetic CA by coordinating the zinc ion with azoles or derivatives. For example [16,17], the Zn-dimethyl-1,2,3-benzotriazole (Me_2bta) and $Zn(OAc)_2{\cdot}H_2O$ show high CA-mimetic catalytic activity, and Zn-based MOFs as mimetic CA demonstrated high solvent/thermal stability and high reusability. Nevertheless, in all the available work, the synthetic process is very complicated, the synthetic conditions are harsh, and the used ligands are also costly. Additionally, all the reported mimetic CA materials are solid substances and unsolvable in the aqueous solution, indicating that it can

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only catalyze CO_2 hydration in a heterogeneous manner. Thus, these mimetic CA cannot provide flexible active-sites as natural CA does, and only partial active-sites can be exposed to substrates, subsequently resulting in a low conversion rate. In contrast, liquid mimetic CA with a flexible active-site is desirable. Histidine in the active-site of natural CA, containing imidazole ring, amino, and carboxylic group, is actually different from azoles, even though various azole-based ligands were attempted to synthesize mimetic CA and showed a promising catalytic performance. To maintain exactly the same active-site of natural CA, it will be preferable to use histidine directly as a ligand for synthesizing the mimetic CA. Moreover, histidine is a natural product that is non-toxic, biodegradable, and environmentally friendly. To the best of our knowledge, starting from histidine directly to synthesize mimetic CA as a liquid phase has not been available.

In this work, for the first time, histidine was used directly as the key component to synthesize mimetic CA. To develop a stable liquid-state mimetic CA, the characteristics of deep eutectic solvents (DESs) was incorporated in synthesizing, where histidine as hydrogen bond acceptors (HBA) and glycerol as hydrogen bond donor (HBD), achieving facile preparation without purification, unique properties of high thermal and chemical stability and extremely low vapor pressure [18]. More specifically, CA mimic was prepared by directly mixing zinc chloride ($ZnCl_2$), histidine (His), and glycerol (Gly), in which Zn atom possesses a coordination environment similar to the CA active-site center. The activity of mimetic CA, ZnHisGly, including the effects of pH and temperature, as well as its catalytic kinetics at various pH were studied by catalyzing the hydrolysis of *p*-nitrophenyl acetate (*p*-NPA). The application of ZnHisGly for CO_2 hydration and conversion was investigated. The control experiments, in the presence of HisGly, $ZnCl_2$, glycerol, and histidine, were separately conducted for comparison.

2. Experimental section

2.1. Materials

L-Histidine (99%), glycerol (99%), zinc chloride, *p*-nitrophenyl acetate (*p*-NPA), *p*-nitrophenol, and calcium chloride were all purchased from Sigma-Aldrich. CO_2 gas (>99.5%) in a cylinder was purchased from AGA A/S (Sweden). All chemicals and reagents used in this study are analytical grades.

2.2. Synthesizing DESs of ZnHisGly and HisGly

This work focused on ZnHisGly as CA mimic, while to verify the significance of the Zn coordination environment, a Zn-free DES (HisGly) was also synthesized.

HisGly was synthesized by mixing L-Histidine with glycerol [19]. Briefly, L-Histidine and glycerol were added in a flask with a molar ratio of 1:6, and the mixture was heated at 70 °C until the homogeneous liquid phase formed and dark brown color appeared (around half an hour). The product was dried under vacuum at 60 °C for 2 h, and the product was obtained. Similarly, ZnHisGly was synthesized following the same procedure, while the molar ratio of zinc chloride, L-Histidine, and glycerol was 1:3:18, based on the active site of carbonic anhydrase composition. Finally, the light yellow ZnHisGly and dark brown HisGly were obtained. They are in the liquid state at room temperature and can be stored at ambient temperature.

2.3. Characterization methods

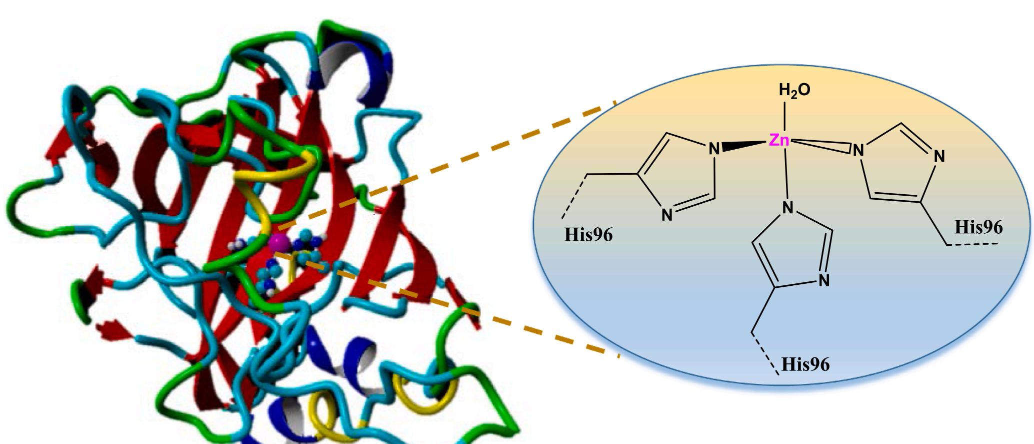
Ultraviolet spectra or analyses were performed by using UV–vis spectrophotometer (UV-1280, Shimadzu). Thermogravimetric analyses (TGA) were performed using a thermal gravimetric analyzer (Netzsch, STA449 F5, Germany). Curves were recorded by heating the samples from 50 to 400 °C under the air atmosphere with a heating rate of 10 °C/min. The freezing point of each DES was determined by a differential scanning calorimeter (DSC) (Netzsch DSC 200F3, Germany) in a temperature range from −78 to 50 °C with a heating rate of 10 °C/min under nitrogen.

2.4. CA mimetic activity assessment

CA mimetic activities of ZnHisGly, HisGly, $ZnCl_2$ were determined by the colorimetric assay based on the hydrolytic reaction of *p*-NPA at room temperature (25 °C), where the product *p*-nitrophenol (*p*-NP) can be monitored at UV402 nm. In a typical experiment, *p*-NPA in acetonitrile (0.2 ml) and the corresponding catalyst (1 mg/ml) in 10 mM PBS buffer (0.2 ml) were successively added to 4.6 ml PBS buffer (10 mM). After the substrate and homogeneous catalyst were thoroughly mixed, the accumulation of *p*-NP was recorded based on the changes of UV absorbance at 402 nm. To eliminate the effect of *p*-NPA self-decomposition, a control experiment was performed without the addition of catalysts. A calibration curve of standard *p*-NP (0–100 μM) in 10 mM PBS buffer (pH = 7.0) was established for evaluating the concentration of the yielded *p*-NP.

2.5. Effects of pH and temperature on CA mimic activity of ZnHisGly

The hydrolysis rate of *p*-NPA, with ZnHisGly (1 mg/ml) or without ZnHisGly, was investigated in the pH from 6 to 10 and at 25 °C, where different pH solutions were prepared by 10 mM NaOH or 10 mM phosphoric acid titration. The product *p*-NP concentration was determined by a UV–vis spectrometer. The hydrolysis rate and activity were



Scheme 1. Structure and active center of natural CA (hCAII, PDB entry: 1H9Q).

calculated, where the activity of ZnHisGly, at pH 7, was set 100% as a reference. Similarly, the hydrolysis rate of *p*-NPA at pH 7, with ZnHisGly (1 mg/ml) or without ZnHisGly, was measured at temperatures from 25 to 80 °C, and the product *p*-NP concentration was determined by UV–vis spectrometer. The hydrolysis rate and activity were calculated, where the activity of ZnHisGly at 25 °C was set 100% as a reference.

2.6. Kinetic evaluation

The kinetic parameters (V_{max}, K_m, and k_{cat}) of ZnHisGly towards the hydrolysis of p-NPA were determined at various concentrations of the substrate ranging from 0.25 to10 mM at room temperature (25 °C). The Michaelis-Menten constant (K_m) and maximum velocity (V_{max}) values were estimated based on the Lineweaver-Burk plots as follows:

$$\frac{1}{V} = \frac{K_m}{V_{max}}\frac{1}{[S]} + \frac{1}{V_{max}}$$

where [S] is the substrate concentration, K_m is the Michaelis–Menten constant, and V_{max} is the maximum reaction velocity. k_{cat} is the enzymatic catalytic rate and is defined as the number of substrate molecules that are converted to the product per each enzyme per unit time, and k_{cat}/K_m reflects the catalytic efficiency.

2.7. CO_2 hydration and conversion testing

To evaluate CO_2 hydration abilities, the commercially available CO_2 gas stream at a rate of 30 ml/min was bubbled into 30 ml Tris-HCl buffer (pH = 8.0, 50 mM) at room temperature (25 °C) containing 1 mg/ml catalysts. The pH values resulting from the generation of H^+ were recorded every 5 min for 30 min. The blank experiment was conducted in the absence of catalysts. For CO_2 conversion, the excessive $CaCl_2$ solution (2 M) was put into the supernatant after the hydration process. The mixture was stirred for around 1.0 h and then let it stand to precipitate $CaCO_3$. The yielded $CaCO_3$ was collected by centrifugation, washed with water, and dried at 60 °C overnight under a vacuum. The mass of $CaCO_3$ was finally weighed and recorded.

3. Results and discussion

3.1. Characterization of ZnHisGly and HisGly

As the freezing point of DES, lower than either of the individual components, was a key criterion to identify DES [20], the freezing points of ZnHisGly and HisGly were characterized by DSC to confirm that both were DESs. As shown in Fig. 1, the freezing point of ZnHisGly was −75 °C, which is lower than those of $ZnCl_2$, His, and Gly, indicating the hydrogen bonding network was built between zinc ion, histidine, and glycerol [19]. Likewise, HisGly is kept as liquid down to −78 °C, indicating that its freezing point is lower than that of ZnHisGly. Therefore, ZnHisGly and HisGly are confirmed to be DESs.

The thermal stabilities of ZnHisGly and HisGly were investigated by TGA, as shown in Fig. 2. There is no obvious weight loss before 150 °C for both DESs, indicating their high thermal stability. Moreover, compared to HisGly, ZnHisGly containing zinc has better thermal stability. Taking the mass loss of 50% as an example, the corresponding temperatures of ZnHisGly and HisGly were 240 and 201 °C, respectively.

To demonstrate that the mimetic structure of ZnHisGly maintains the active-site of CA, ZnHisGly was inspected by (1) the coordination of zinc to histidine, and (2) the ratio of zinc vs histidine (if the ratio is 1:3). First, both ZnHisGly and HisGly were characterized by UV–vis spectrometer, as shown in Fig. 2b. The absorption peak of ZnHisGly has a blue-shift compared to that of HisGly, proving the coordination of zinc ion and HisGly. Second, when the molar ratio of zinc ion vs histidine was 1:3, a solution was formed, indicating that ZnHisGly could be synthesized with the ratio of 1:3, being consistent with the active site of CA composition (molar ratio of zinc ion vs HisGly is 1:3).

All these results demonstrated that ZnHisGly was a DES with a mimetic structure maintaining the active-site of CA.

3.2. CA mimetic activity of ZnHisGly

Hydrolysis of *p*-NPA is typically served as a model reaction to evaluate the activities of CA and its mimics in terms of the similar mechanism between CO_2 hydration and the hydrolysis of *p*-NPA [21,22]. Therefore, the CA mimetic activity was evaluated and quantified based on the generation rate of the hydrolyzed product *p*-nitrophenol (*p*-NP) via detecting the intensity of its characteristic absorption peak at 402 nm (Fig. 3). Since *p*-NPA has self-decomposition in the absence of catalyst, which was set as control, in this part, six systems, i.e., ZnHisGly, HisGly, $ZnCl_2$, glycerol, histidine, and control, were investigated. As presented in Fig. 4, absorption of *p*-NP in the presence of ZnHisGly increased with increasing time, and the concentration of *p*-NP reached 50 μM at 30 min, showing the best catalytic performance. In the absence of zinc ion, CA mimetic activity of HisGly decreased significantly, and the concentration *p*-NP was only 12.8 μM at 30 min, indicating that the zinc ion as active central site played a decisive role in catalytic activity. According to the reported mechanism of CA [23], it was revealed that the zinc ion was coordinated with three histidine molecules and acted as the active site, where zinc ion launched a nucleophilic attack toward CO_2 and subsequent generation of the bicarbonate ion. This explains the role of zinc ion, being consistent with the observation in this part.

Also, as shown in Fig. 4, these raw materials for ZnHisGly synthesis

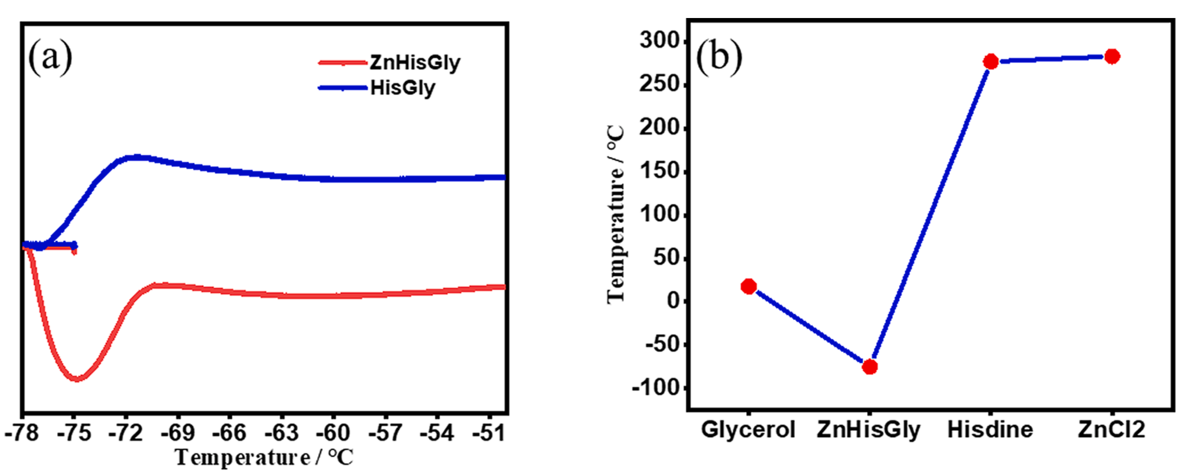


Fig. 1. (a) DSC curve of ZnHisGly and HisGly, and (b) freezing points of ZnHisGly, Gly, His, and $ZnCl_2$.

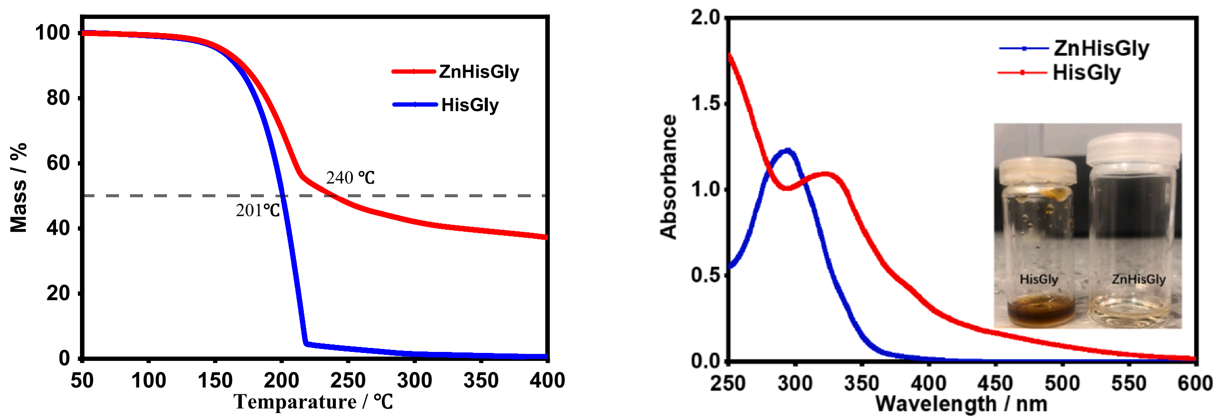
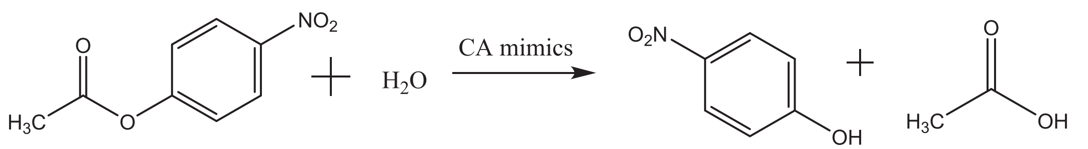


Fig. 2. (a) TGA curve of ZnHisGly and HisGly; (b) UV curve of ZnHisGly and HisGly.



p-NPA *p*- NP (402 nm)

Fig. 3. Hydrolysis reaction of p-NPA in the presence of CA mimics.

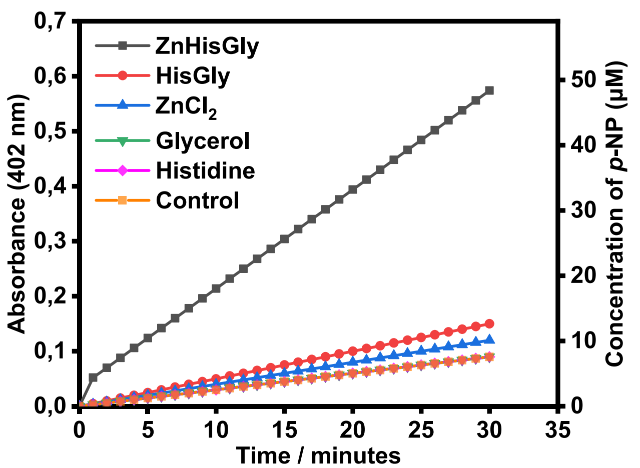


Fig. 4. Time course of optical absorbance (402 nm) during hydrolysis of *p*-NPA to *p*-NP.

showed a relatively lower activity on the hydrolysis of *p*-NPA. Particularly, glycerol and histidine showed exactly the same activity as the control, indicating there is no catalytic activity at all. After building an analogous active-site of CA by using these raw materials, the activity of ZnHisGly was significantly improved and presented an efficiently catalytic performance.

3.3. Effects of pH and temperature on CA mimic activity of ZnHisGly

It is well-known that natural enzyme CA is sensitive to changes in physical and chemical conditions and is easily denatured by even a slight increase in temperature or change of pH. To demonstrate the advantages of mimetic ZnHisGly, its activity at different pH (6–10) and temperatures (25–80 °C) was investigated, where that at pH = 7 and 25 °C was used as a reference, i.e., 100%.

The pH influence on the mimetic activity of ZnHisGly is shown in Fig. 5(a), together with that of the control system. The hydrolysis rate of *p*-NPA by ZnHisGly increased sharply as the increase of pH, and that for the system of control behaves similarly. This indicates that the more alkaline solution was, the deeper the degree of hydrolysis was, implying

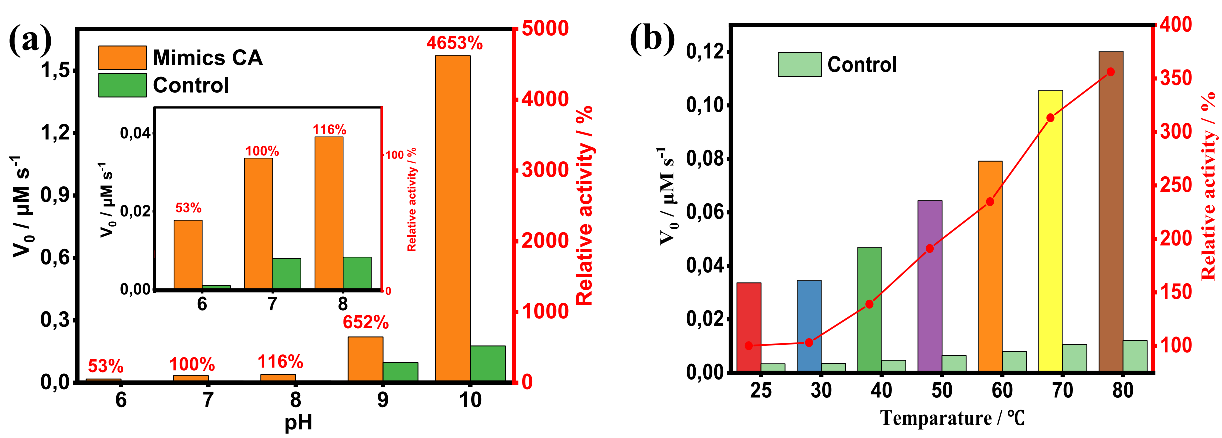


Fig. 5. Catalyst activity with and without ZnHisGly at different (a) pH values, and (b) temperatures.

alkalinity could contribute to the hydrolysis of *p*-NPA and enhance the activity of ZnHisGly [24]. It is probably because that alkaline can neutralize the hydrolyzed product, acetic acid, accelerating the reaction move forward. Notably, much higher activities of ZnHisGly than the control system were observed, indicating ZnHisGly can also greatly promote the hydrolysis of *p*-NPA. Specifically, the relative activity increased to 4653% at pH = 10, which is about 88 times higher than that at pH = 6. According to the research by Jonathan et al. [23], the rate-limiting step of hydration reaction in CA mimic is the deprotonation of the bound water molecule from zinc ion. Therefore, the alkali environment will facilitate deprotonation of the bound water molecule from zinc ion and thus enhance CO_2 hydration, being consistent with the observation in the experiment. Indeed, Kim et al. [21] synthesized CA mimetic materials with different anions in zinc-based salts and evaluated their catalytic performance. It was observed, among the Zn salts with different anions, $Zn(OH)_2$, i.e., the functionalized zinc ion with a hydroxyl group, was used directly to achieve deprotonation of the bound water molecule from zinc ion, and showed the highest catalytic performance with increased hydrolysis activity. This implies the rate-limited step for mimetic CA is the process of deprotonation, and skipping the rate-limited step can indeed enhance the conversion rate. Therefore, the combination of ZnHisGly and hydroxyl ions resulted in desirable catalytic activities at high pH.

Fig. 5(b) showed the hydrolysis rate and relative activities of ZnHisGly at pH = 7 under various temperatures. Hydrolysis rate and relative activities increased gradually with the increase of temperatures, probably owing to the acceleration of mass transfer [25]. The best performance for the hydrolysis of *p*-NPA was observed at 80 °C with a relative activity of 360%, indicating the mimetic activity of ZnHisGly increased with increasing temperature and has the advantage over natural CA on temperature.

All in all, an increase in pH and temperature could remarkably enhance the mimetic activity of ZnHisGly, which has distinct advantages over characteristics of nature CA and provides the possibility to catalyze reactions under harsh conditions. While natural CA shows the highest activity at pH 7.5 and temperature 37 °C [26], and the decrease of CA activity is unavoidable as protons are generated along with the CO_2 hydration, leading to the decrease of pH in solution and thus the decrease of CA activity. Therefore, the application of mimetic CA can address this issue and also be applicable under high-temperature conditions, which can broaden the application of CA by unlocking the restriction of natural CA.

Based on the results above, the pH value in the solution plays a more important role in improving the activity of ZnHisGly. Therefore, the kinetic behaviors of ZnHisGly in the pH range from 6 to 10 were evaluated. Since Michaelis–Menten kinetics was typically used for studying the carbonic anhydrase kinetics [27] and this study was to mainly develop carbonic anhydrase mimetic materials, the Michaelis–Menten model was chosen for describing the mimetic enzymatic kinetics. As a result, the mimetic enzymatic kinetics could be fitted very well (correlation coefficient: 0.97) by using the Michaelis-Menten model. The corresponding kinetic parameters, including maximum velocity (V_{max}), Michaelis constant (K_m), catalytic constant (k_{cat}), and catalytic efficiency (k_{cat}/K_m), are listed in Table 1. K_m is the substrate concentration at which the reaction rate is at half-maximum, and lower K_m means a stronger affinity and results in a higher conversion rate of reaction. k_{cat}/K_m is a measure of the efficiency of an enzyme for converting a substrate into product. It was found that K_m values decreased as the increase of pH, implying affinity between mimetic ZnHisGly and substrate was enhanced, and the activity of mimetic ZnHisGly was strengthened. Consequently, as the affinity increased with the increase of pH, the catalytic constant k_{cat} was increased, representing the enhanced conversion rate of reaction. Finally, the catalytic efficiency of ZnHisGly was calculated based on the results of K_m and k_{cat}. At pH = 10, the highest k_{cat}/K_m was obtained with 25198 $M^{-1}s^{-1}$, which is about 21,988 times higher than that of pH = 7. These results revealed that pH plays a critical role in enhancing the activity of mimetic ZnHisGly and its activity is unrestricted by pH, which is also consistent with the results of CA mimic activity at different pH.

Table 1
Kinetic parameters for the hydrolysis of *p*-NPA in the presence of ZnHisGly at different pH.

pH	V_{max} (mM/min)	K_m (mM)	K_{cat} ($\times 10^{-3}$ s^{-1})	K_{cat}/K_m ($M^{-1}s^{-1}$)
6	0.513	77.63	12.42	0.160
7	0.991	21.28	24.39	1.146
8	1.149	14.82	28.48	1.921
9	1.471	2.037	164.6	80.85
10	8.957	0.047	1175	25,198

A comparison of catalytic efficiency (k_{cat}/K_m) of mimetic CA to that of natural CA (hCA II) allows for a better identification how the development of mimetic CA is. The maximum catalytic efficiency of hCA II is 1.5×10^8 $M^{-1}s^{-1}$ [28], while the catalytic efficiency of ZnHisGly is 2.5 $\times$ 10^4 at pH = 10. Therefore, the catalytic efficiency of ZnHisGly is 3.7 orders of magnitude lower than that of natural CA. Base on the investigation of mimetic CA at different pH and temperatures, higher catalytic efficiency will be obtained in the stronger alkaline solution and at high temperature, for example, the relative activity was increased 7-fold when pH increased from 9 to 10 and also increased 3-fold when temperature increased from 25 to 80 °C. Therefore, ZnHisGly has an opportunity to approach the activity of natural CA by adjusting conditions, even better than CA. Additionally, to the best of our knowledge, the previously developed catalytic efficiency of CA mimics (i.e., the solid CA mimic materials) are always around 4–5 order of magnitude slower than that of the natural enzyme [23,29,30], and thus ZnHisGly developed in this work greatly improved the activity. Moreover, ZnHisGly was more applicable and flexible than those solid CA mimic materials, for instance, it could be potentially immobilized in the support materials, e.g., membrane, carbon materials, for versatile application.

3.4. CO_2 hydration and conversion

In this study, DES ZnHisGly was used as the mimetic catalyst (1 mg/ml) for CO_2 conversion rather than the solvent for CO_2 capture. The CO_2 hydration and sequestration by ZnHisGly were then conducted for illustrating their CO_2 sequestration ability. In this regard, the ability of ZnHisGly for CO_2 hydration and conversion (sequestration) was examined and also compared with HisGly and control. Theoretically, the hydration of CO_2 by mimetic CA in the aqueous solution could produce HCO_3^- and H^+, resulting in a decrease of pH value. The lower pH of the solution was obtained, indicating that more protons were generated under the catalysis of mimetic CA. As shown in Fig. 6(a), the pH values decrease over time as bubbling CO_2 in aqueous solution and the pH changes followed the order of ZnHisGly > HisGly > control, which is consistent with the results for the hydrolysis of *p*-NPA. In the presence of ZnHisGly, pH decreased from 7.0 to 6.2 within 10 min, which showed the biggest drop of pH compared to others, indicating that ZnHisGly could significantly accelerate the hydration of CO_2. Additionally, CO_2 sequestration was further examined by adding calcium chloride. Based on the mechanism shown in Fig. 6b, HCO_3^- produced from the hydration of CO_2 could be sequestrated as $CaCO_3$ precipitates by reacting with excessive $CaCl_2$. The weights of obtained $CaCO_3$, in the presence of ZnHisGly, HisGly, and control, were 100, 60, and 32 mg, respectively. As expected, for the solution in the presence of ZnHisGly, the amount of the generated $CaCO_3$ was the most among the studied systems, confirming ZnHisGly indeed promoted the hydration and sequestration of CO_2. All these results demonstrated that the development of analogous active-site of CA, i.e., ZnHisGly, presented outstanding catalytic performance of CO_2 hydration and sequestration. Also, CA mimetic ZnHisGly is a promising alternative to natural CA for CO_2 sequestration.

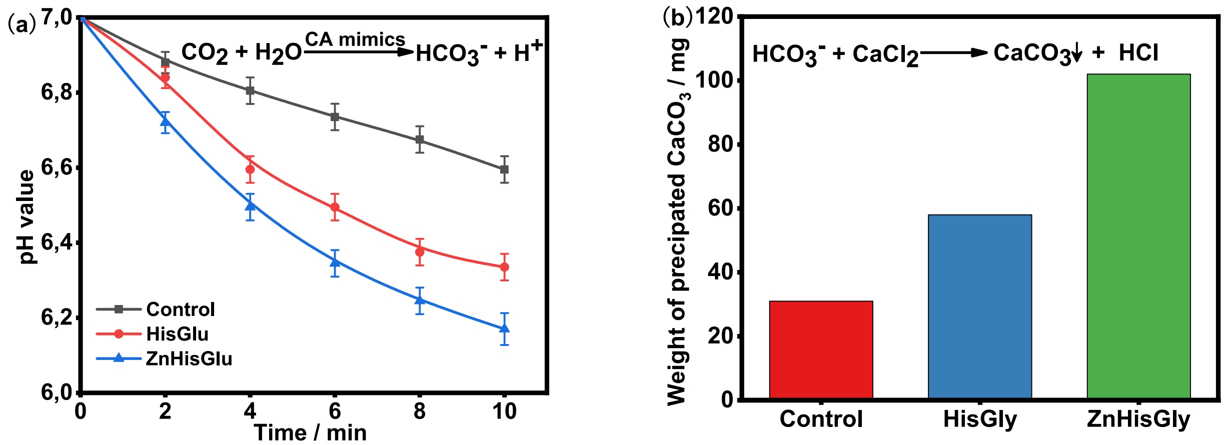


Fig. 6. (a) pH changes during hydration of CO_2, and (b) weights of formed $CaCO_3$ during sequestration in the presence of different catalysts.

4. Conclusions

In this work, a Zn-containing DES (ZnHisGly) that possess similar active sites as CA was developed, and its mimic activity was evaluated and compared with HisGly, $ZnCl_2$, glycerol, histidine, and control. ZnHisGly can be easily prepared by one-step route without any purification and had good thermal stability. Investigation for the hydrolysis of *p*-NPA confirmed that ZnHisGly has desirable CA mimic activities with a high *p*-NP concentration of 50 μM for 30 min at pH 7 and 25 °C, which is much higher than those of HisGly, $ZnCl_2$, glycerol, histidine, and control. Notably, ZnHisGly exhibited much better catalytic performance at high pH and temperatures. In addition, ZnHisGly could promote the hydration and sequestration of CO_2, which can be integrated or coupled with CO_2 capture/separation and purification. This study provides an efficient way to design CA mimic for CO_2 capture/separation, purification, and conversion.

CRediT authorship contribution statement

Zhibo Zhang: Methodology, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Fangfang Li:** . **Yi Nie:** Conceptualization. **Xiangping Zhang:** Conceptualization. **Suojiang Zhang:** Supervision, Conceptualization. **Xiaoyan Ji:** Conceptualization, Supervision, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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